

ADITYA ROY BARDHAN

Kolkata, India

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ML Engineer

ML Engineer specializing in hyperscale LLM training and AI systems, with hands-on production experience on a 519B-parameter model. Unique combination of deep model-training expertise — 3D parallelism, DeepSpeed, reasoning-trace construction — and a rigorous cybersecurity background securing international financial infrastructure. Driven by the intersection of frontier AI systems and real-world data quality, with a track record of owning end-to-end pipelines from raw data ingestion to production model deployment.

TECHNICAL SKILLS

LLM Training
ML / AI
Data
Systems
Security

Links

[GitHub](#)
[LeetCode](#)
[Kaggle](#)
[LinkedIn](#)
[Instagram](#)
[portfolio](#)

PROFESSIONAL EXPERIENCE

1 ML Training Engineer SK TELECOM

Dec 2025 — Feb 2026

- Owned end-to-end training cycles for a 519B-parameter production LLM — one of the largest models in deployment in East Asia.
- Implemented 3D parallelism (Data Parallelism, Pipeline Parallelism, Tensor Parallelism) via Megatron-LM across multi-node GPU clusters, sustaining high Model FLOP Utilization (MFU) at scale.

2 Freelance Security Consultant

INDEPENDENT CYBERSECURITY ADVISOR

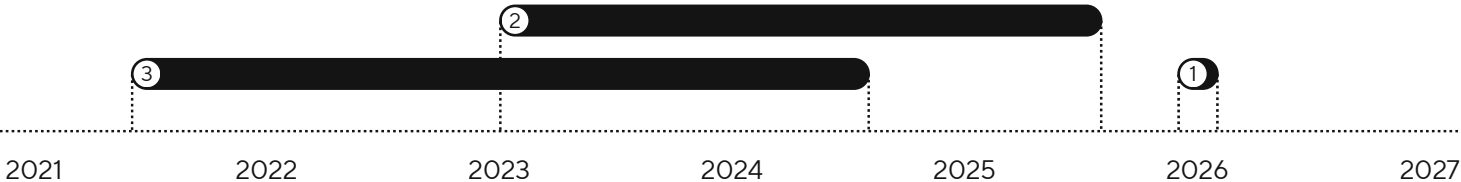
Jan 2023 — Aug 2025

- Held legal security mandates with 8 international banks, conducting API penetration testing, threat modeling, and mitigation planning for systems processing multi-million dollar transactions daily.
- Designed anomaly detection and alerting systems for real-time

3 Technical Operations RIOT GAMES

Jun 2021 — Aug 2024

- Maintained backend service stability and performance for high-concurrency gaming infrastructure, specializing in low-latency synchronization across cross-regional server clusters.



- Integrated DeepSpeed ZeRO-3 for optimizer state sharding and FlashAttention-2 for memory-efficient attention — reducing peak

VRAM consumption and enabling larger effective batch sizes.

- Constructed and curated reasoning-trace datasets to be used in knowledge distillation — transferring capabilities from the 500B+

teacher to smaller student models while preserving chain-of-thought integrity. This work gave direct exposure to how annotation quality affects distillation fidelity.

- Diagnosed and fixed training instabilities (loss spikes, gradient explosions) by auditing data pipelines — confirming that malformed or

low-diversity batches were the root cause in most incidents.

- Collaborated cross-functionally with data, infra, and alignment teams on prompt engineering standards and output evaluation

rubrics to feed back into the data labeling pipeline.

financial transaction monitoring — experience directly transferable to data quality assurance pipelines.

- Delivered detailed vulnerability reports and remediation roadmaps; advised on encryption, network segmentation, and identity

verification infrastructure.

- Worked on observability tooling, performance profiling, and incident response for distributed systems serving millions of concurrent

users.

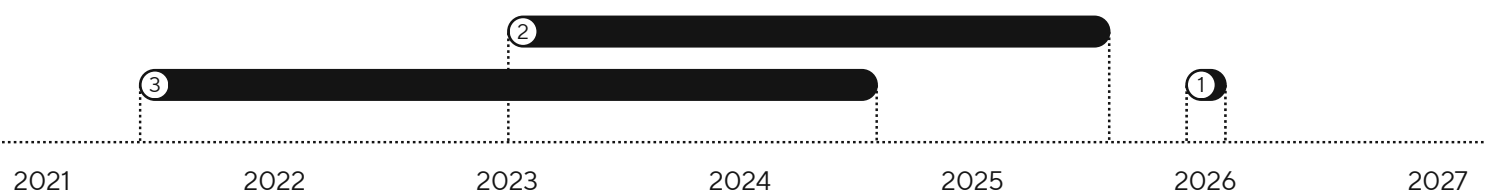
TECHNICAL PROJECTS & RESEARCH

ASHA — Alerting Satellite Heat Analytics (GitHub)

- Built a production-grade real-time satellite fire detection and heat analytics system for the entire Indian subcontinent — covering 8.4°N–35.0°N latitude, 68.7°E–97.4°E longitude. Integrated NASA FIRMS (Fire Information for Resource Management System) and NDMA data to monitor 7 fire-prone regions: Northeast India, Central India (Chhattisgarh/MP), Odisha, Andhra Pradesh & Telangana, Karnataka, Western Ghats, and the Himalayas. Engineered confidence-scored fire event classification, regional fire intensity analytics, and an interactive geospatial map centered at 21.7°N, 83.0°E for field responders and disaster management agencies. System outputs real-time alerts and analytics dashboards built in HTML/JS — designed for integration with NDMA emergency response workflows.

Satellite Imagery for Global Change Detection (GitHub)

Built a pixel-level segmentation model to detect wildfires, smoke plumes, and atmospheric anomalies from geostationary satellite feeds using NASA Earthdata APIs. Developed a custom geospatial annotation pipeline to label multi-spectral imagery — hands-on experience building ground-truth datasets for vision models at scale. Implemented data cleaning and quality filtering heuristics on heterogeneous, noisy satellite data — significantly improving mIoU on held-out test sets.



Real-Time Demand Forecasting & Price Optimization (GitHub)

Built high-frequency demand forecasting models for Uber ride requests and Walmart-style inventory scenarios – incorporating geospatial dependencies, temporal lag features, and external demand signals. Engineered 200+ features from raw transactional and geospatial data; built automated validation pipelines to catch distribution shifts and data drift before retraining. Designed a dynamic pricing algorithm that responds to predicted demand surges – achieving sub-100ms inference via model quantization and ONNX export.

J.A.R.V.I.S. – Multimodal AI Assistant

Designed and built an end-to-end multimodal AI assistant processing voice, text, and visual inputs – integrating speech recognition, NLP pipelines, and computer vision into a unified inference stack. Implemented custom intent classification and entity extraction for task automation, system-level control, and multi-API orchestration.

Aadhaar Pravah – Identity Fraud Detection

One of India's largest national-level hackathons, competing against thousands of teams. Designed a real-time data flow analyzer for Aadhaar-scale identity streams to detect fraud patterns and data integrity violations with low-latency alerting. Built and labeled a synthetic fraud dataset with deliberate edge-case construction to improve recall on rare fraud events; implemented AES-256 encryption and role-based auditing for compliance.

Ancient Indian Yantra Reconstruction Engine

Grand Finalist at Smart India Hackathon 2025 – Building a geometric computation engine to algorithmically reconstruct and visualize ancient Indian astronomical Yantras – applying graph algorithms, symbolic math, and constraint solving to historically ambiguous source material.

EDUCATION

Department of Data Science & Artificial Intelligence

Indian Institute of Technology (IIT), Guwahati

,Coursework covering machine learning theory, statistical inference, deep learning, NLP, computer vision, and distributed systems. ,Active participation in SIH 2025 (Grand Finalist), hackathons, and open-source ML projects via GitHub Pro.

RECOGNITION & INTERESTS

SIH 2025 Grand Finalist

Smart India Hackathon 2025 – National-level competition with thousands of teams. Recognized for "Siddhant" – Ancient Indian Yantra Reconstruction Engine.

GitHub Pro

Active open-source contributor – github.com/aditya2909rb.

LeetCode

leetcode.com/u/adityaroybardhan2909 – consistent problem solving in algorithms and data structures.



Algorithm Design

Advanced Rubik's Cube (4x4, 5x5)
— group theory applied to physical
optimization.

Computational History

Reconstructing ancient Indian
astronomical Yantras as geometric
computation problems.

